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Alpha-Spectroscopy Analysis for Quality Control of ^{225}Ac Radiopharmaceuticals

Research Project Report

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Abstract

This report sets out the results of a summer research project on the analysis of alpha-particle spectra for quality-control measurements of ^{225}Ac radiopharmaceuticals. Using an existing multi-isotope fitting framework developed within the research group, spectra acquired using thin-layer chromatography (TLC) were analysed to study the stability of the fitting procedure, physically and empirically tighten parameter bounds and investigate correlations between fitted parameters. The resulting procedure was applied to a 10-day radiolabelling study and a later unlabelled dataset. A toy Monte Carlo study was also developed to test the recovery of known parameters and the calibration of their fitted uncertainties. This work identified important parameter degeneracies and biases in parts of the fitting procedure, while the tighter bounds improved the precision and consistency of the fitted results. The study also established a framework for further validation of the fitting method and for future analysis of alpha-TLC data. In parallel, several silicon photodiodes were tested experimentally using ^{241}Am and ^{210}Po alpha sources.

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Chapter 1

Introduction

1.1 Research context

In the context of nuclear medicine, treatments based on radioactive nuclides have gained extensive interest over recent decades. Some of them use a biological targeting molecule to deliver a radioactive isotope preferentially to cancer cells. Alpha particles are of particular interest because of their high linear energy transfer and short range in tissue, which allow a large amount of energy to be deposited locally while limiting irradiation of surrounding tissue [1, 2]. A good candidate for this purpose is ^{225}Ac . Its half-life of 9.9d allows for production, radiolabelling, processing and transport before clinical administration, while leaving time for biological targeting inside the body [2, 3].

One challenge in this development is radiolabelling. The radionuclide must be attached to a molecule that can reach the relevant biological target. A chelator¹ is used to bind ^{225}Ac , while the conjugated targeting vector provides the biological selectivity [1, 2]. Quality-control (QC) measurements are therefore required to determine how much of the radionuclide is bound to the radiolabelled compound and how much remains free.

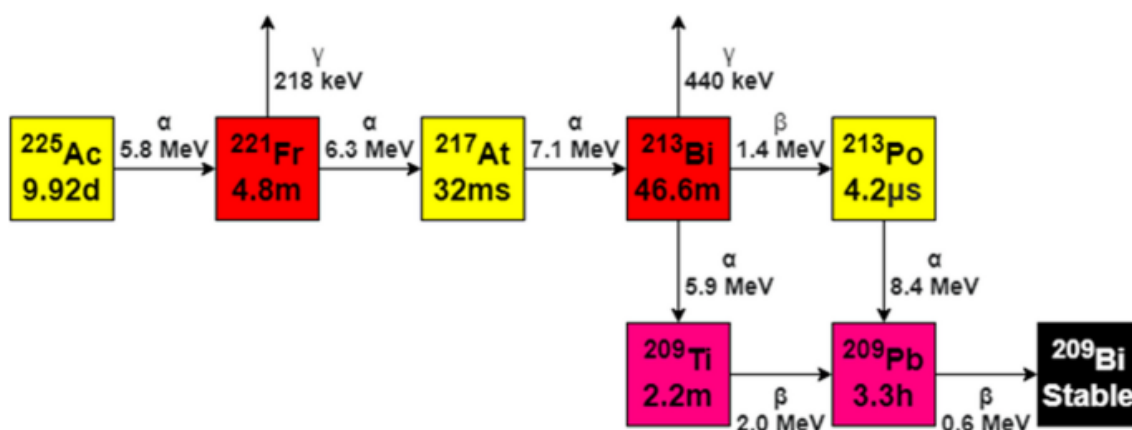


Figure 1.1: Illustration of the decay chain of ^{225}Ac . From Hooijman et al. [1].

This isotope, however, does not have a strong gamma emission suitable for direct QC. Gamma-based measurements can instead use emissions from its daughter isotopes,

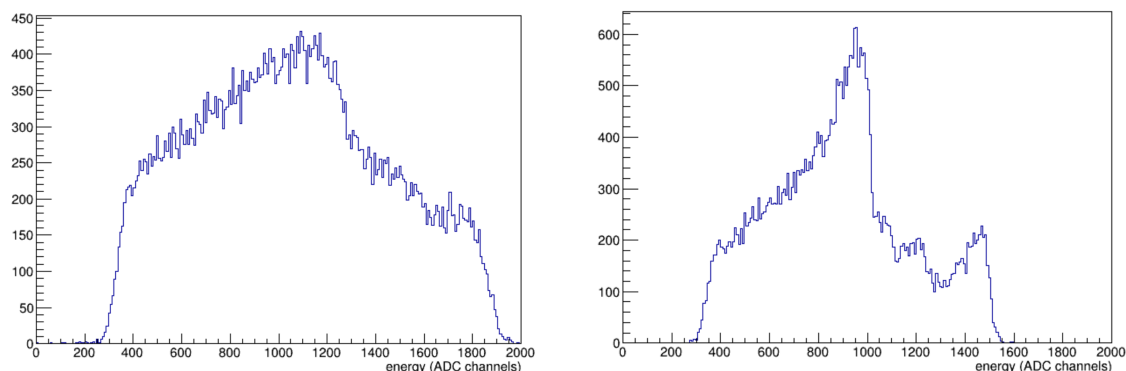
¹A chelator is a molecule with several binding groups that coordinate a metal ion to form a stable complex. In a targeted radiopharmaceutical, it can be conjugated directly or through a linker to a biological targeting vector.

which requires the state of daughter equilibrium to be considered [1]. Direct alpha spectroscopy provides an alternative by allowing ^{225}Ac and its alpha-emitting daughters to be distinguished through the different alpha energies in the decay chain, illustrated in Fig. 1.1.

1.2 Alpha spectra: origin and characteristics

One QC method used to assess the radiolabelling of ^{225}Ac is thin-layer chromatography (TLC), in which a small liquid sample of radiopharmaceutical is spotted onto a TLC strip. The liquid moves through the strip by capillary action and carries the chemical species with it. The free radionuclide and the radiolabelled compound interact differently with the material and therefore migrate by different amounts, effectively becoming separated [3].

The data used in this report were acquired at TRIUMF using TLC. Silicon photodiodes were placed over each end of the TLC strip, receiving alpha particles emitted from it and producing spectra from the free and labelled fractions, respectively. The detected alpha particles lose energy while traversing the TLC material and, in the experimental setup, air and other intervening material. Different path lengths therefore produce broad distributions with step-like high-energy edges rather than narrow lines, and the contributions from the different isotopes overlap as shown in Fig. 1.2 [3].



(a) Spectrum from channel 0 corresponding to the free actinium part.

(b) Spectrum from channel 1 corresponding to the labelled actinium part.

Figure 1.2: Two examples of alpha spectra recorded using TLC, corresponding to each end of the TLC strip.

The latter shows spectra recorded with each of the two silicon detectors, corresponding to the free (Fig. 1.2a) and labelled (Fig. 1.2b) fractions of the solution.

1.3 Objectives

The aim of this project was to fit these spectra to derive activity estimates for ^{225}Ac and its daughters in both the labelled and free fractions. From these results, quantities such as the free-to-total ^{225}Ac fraction, $A_{\text{free}}/(A_{\text{free}} + A_{\text{labelled}})$, could be extracted. The complementary labelled-to-total fraction is used in some graphs later in this report. The fitting model and core analysis code had already been developed within the research group before this project. Much of the analysis of the 10-day study was then carried out collaboratively by a

group of two to four undergraduate researchers, including myself, who divided the spectra and compared fitting strategies and parameter constraints. My individual work included fitting my assigned datasets, investigating parameter correlations and model behaviour, developing the toy Monte Carlo validation, analysing part of the later unlabelled dataset and contributing to the detector tests.

The fitting model was complex, so the shared aim of the analysis was to study its effectiveness, precision and weaknesses with the goal of optimising the fitting procedure and extracting reliable information with it.

Chapter 2

Alpha spectroscopy fitting

2.1 The fitting model

The measured spectra were fitted using a model based on the expected step-like response of each alpha emission. Each isotope in the ^{225}Ac decay chain was represented by a sum of smooth step functions:

$$f(x) = a \max \left(0, \min \left(1, 1 + \frac{c(x - x_0)}{x_0} \right) \right) \left[1 - \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x - x_0}{b\sqrt{2}} \right) \right) \right], \quad (2.1)$$

where x is the ADC channel, x_0 is the position of the high-energy step edge, a is the step amplitude, b controls the Gaussian smearing of the edge and c controls the low-energy slope. Each isotope contains several alpha-emission lines, so its contribution is a sum of smooth steps whose relative positions and amplitudes are set using known alpha energies and emission probabilities [4]. The full spectrum was obtained by summing the contributions from the alpha-emitting isotopes in the decay chain: ^{225}Ac , ^{221}Fr , ^{217}At , ^{213}Bi and ^{213}Po .

Physical constraints from the decay chain were used to limit the number of free parameters. Only $x_{0,\text{Ac}225}$ and $x_{0,\text{Po}213}$ were independent position parameters; the positions of ^{221}Fr , ^{217}At and ^{213}Bi were derived from them using the known energy ratios. The ^{221}Fr amplitude was linked to that of ^{217}At through their stopping-power ratio, while the ^{213}Bi alpha contribution was constrained by its small alpha branching fraction. The remaining parameters were determined by a weighted least-squares fit to the measured spectrum.

In the one-stage fitting routine used for most of this work, the smooth-step amplitude component of isotope i , a_i , was converted to an activity estimate according to

$$A_{\text{step},i} = 2S_i a_i, \quad (2.2)$$

where S_i is the alpha-particle stopping power in air [5].

As Fig. 1.2b shows, channel 1 runs presented additional peaks near the high-energy edges of some isotope contributions. The origin of these peaks has been attributed by the project team to a significant fraction of solution containing labelled ^{225}Ac being deposited on the top surface of the TLC strip, which faces the detector directly. Alpha particles emitted from this surface component avoid most of the energy loss associated with traversing the TLC material, so they appear close to the high-energy edge and form a narrower feature than the smooth-step contribution. To model this, an optional Crystal Ball component

was added to the smooth-step response of isotopes that presented these peaks. The Crystal Ball function consists of a Gaussian core joined to a low-energy power-law tail and is described by an amplitude a_{cb} , a transition parameter β , a power-law exponent m and a Gaussian-core width w . When a Crystal Ball component was enabled, the routine applied the same conversion to its fitted amplitude, $A_{cb,i} = 2S_i a_{cb,i}$, and used $A_i = A_{step,i} + A_{cb,i}$ in the activity ratios.

2.2 10-day TLC study

The first dataset we had to fit was a 10-day study carried out at TRIUMF in which radiolabelling performance was assessed on successive days after ^{225}Ac purification using different labelling systems and concentrations. On each study day, an aliquot from the stored ^{225}Ac stock was used to start a new radiolabelling reaction. For each labelling system, aliquots were then taken from the same continuing reaction at the times shown in Table 2.1, spotted onto a TLC strip, developed and measured for five minutes. The day number therefore represents the time since ^{225}Ac purification, whereas the listed times represent the time since the start of that day's radiolabelling reaction.

Table 2.1: Characteristics of the different runs taken each day with each labelling system.

Labelling system	Concentration (M)	1st Run	2nd Run	3rd Run
Ab-DOTA	46×10^{-6}	30 min	60 min	24 h
Macropa	10^{-5}	10 min	20 min	24 h
Macropa	10^{-6}	10 min	20 min	24 h

All available spectra were divided among the undergraduate team, giving priority to the Ab-DOTA runs. I was initially responsible for Macropa 10^{-6} runs and fitted additional runs after completing them. Although the spectra were divided between us, we used the same fitting framework, regularly compared results, discussed problematic spectra and agreed on common strategies for constraining the parameters. An example of the application of the fitting routine to a spectrum is shown in Fig. 2.1.

A value of χ^2/ndf close to unity is commonly used as an indication that the model describes the data within the assigned uncertainties, although it does not by itself guarantee that all fitted parameters are well constrained [6]. Some runs presented anomalous behaviour, and small changes in the initial conditions yielded very different parameter estimates even with χ^2/ndf reasonably close to 1. Parameter b , for example, controls the Gaussian smearing of the high-energy edge and is shared by all isotopes except ^{213}Po , which has its own b_{Po213} . In the aforementioned fit, b was found to be 78.02 ± 141.7 , showing that the parameter was very poorly constrained by that spectrum. This motivated further investigation and a more systematic approach towards fitting.

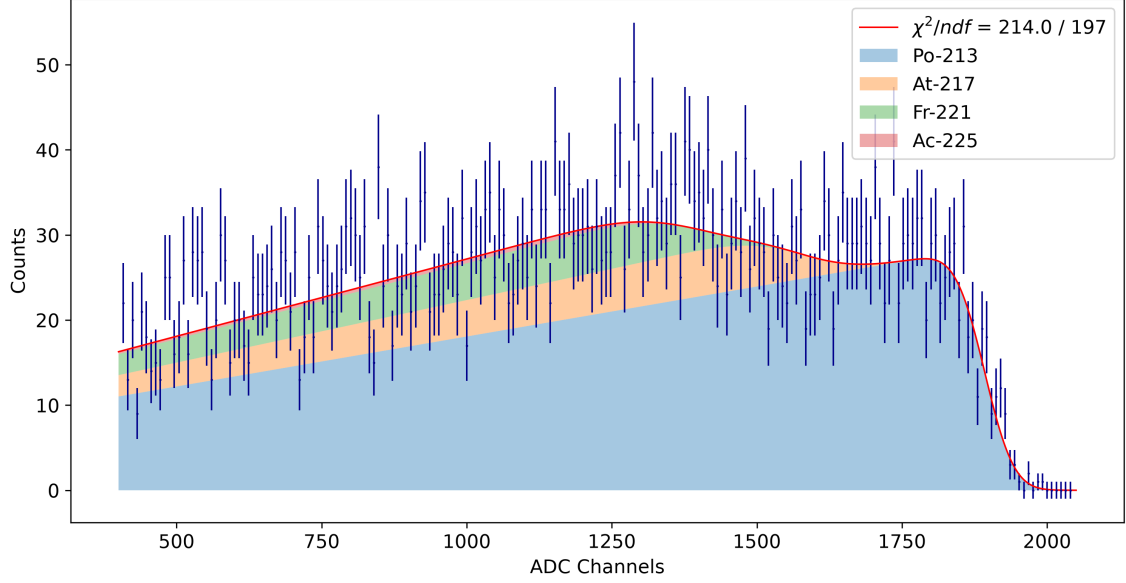


Figure 2.1: Example of a problematic fit from channel 0, showing a low number of counts, and unclear step edges.

Part of the problem was the large number of fitted parameters and the wide parameter space available to them. In the fitting routine, an initial guess and an allowed interval of values were provided for each parameter. After discussing the previous issue with the group, we tightened several bounds using physical considerations, the behaviour of well-resolved spectra and the distributions of preliminary fitted values, while retaining wider bounds for quantities that could not be constrained reliably. These tighter bounds did not reduce the number of free parameters, but limited the parameter space available to the fit. Parameters $x_{0,\text{Ac225}}$ and $x_{0,\text{Po213}}$, for example, correspond to the fitted ADC-channel positions of the high-energy edges associated with each isotope. For spectra recorded with the same ADC scaling and detector configuration, these positions should remain approximately constant and could therefore be constrained significantly.

Constraining $x_{0,\text{Po213}}$ was straightforward, since the position of the step edge from this isotope was easy to identify visually in all fits, as it had the highest energy and did not overlap with other isotopes (see Fig. 2.1). Since the horizontal axis was not calibrated to energy, I used the derived $x_{0,\text{Po213}}$ from all my fits and calculated the error-weighted mean as:

$$\bar{x} = \frac{\sum_{i=1}^N w_i \cdot x_i}{\sum_{i=1}^N w_i} \quad w_i = \frac{1}{\sigma_i^2}, \quad (2.3)$$

where σ_i is the uncertainty of the corresponding estimate of the parameter [6]. These calculations were carried out separately for channel 0 and channel 1 data, since the axes had different scales. The calculated mean, $\bar{x}_{0,\text{Po213}} \simeq 1863$, provided a first estimate for the constraints used in later fits.

Parameter $x_{0,\text{Ac225}}$ was harder to constrain, however. Fig. 2.1 shows why. Due to the

lower ^{225}Ac activity and the low number of overall counts¹, the position of its step edge was difficult to identify in the spectrum. This made visual checks less reliable, and many of the fits were not suitable for estimating the true value of $x_{0,\text{Ac}225}$. Fortunately, knowing the mean value of $x_{0,\text{Po}213}$, I was able to estimate $x_{0,\text{Ac}225}$ using the ratio between the energies of the dominant alpha emissions of ^{225}Ac and ^{213}Po (see Fig. 1.1), approximately 5.8 MeV and 8.4 MeV, respectively. Since the ADC response was known to be linear and its intercept negligible, I used $\bar{x}_{0,\text{Ac}225} \approx \frac{5.8}{8.4} \bar{x}_{0,\text{Po}213} = 1286$ as a first approximation of the parameter. The limitation was that alpha particles with different energies may not have lost the same fraction of energy in the medium, and secondary alpha lines from each isotope may have further interfered with that.

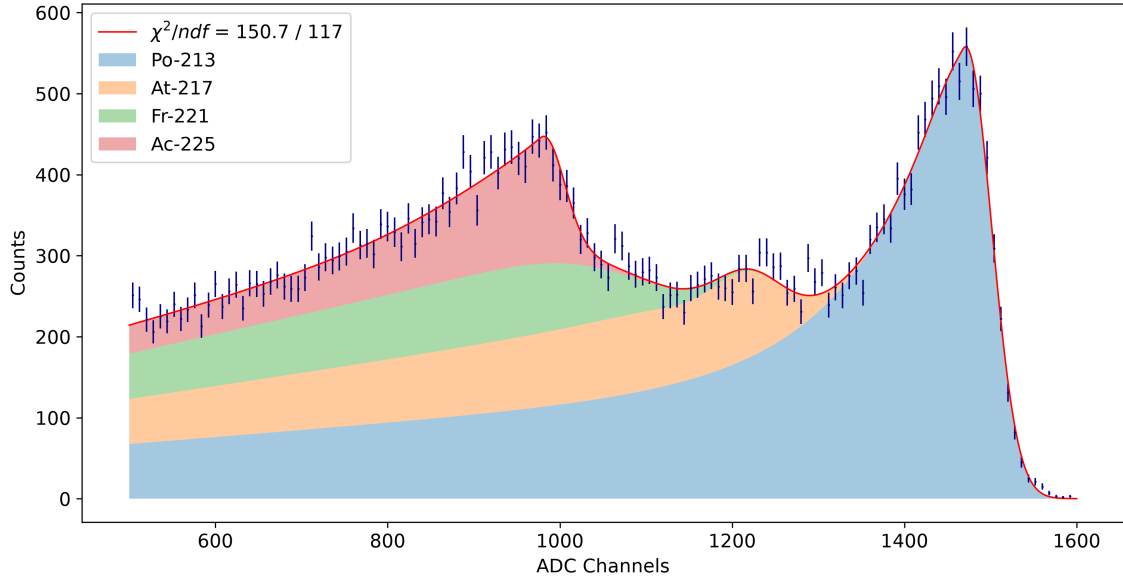


Figure 2.2: Example of a fit from channel 1, presenting a moderate number of counts and Crystal Ball peaks with clear step edges.

Both step edges were very clear in channel 1 data, as Fig. 2.2 shows. This allowed me to relate the ADC scales of both channels using the more precise mean values of $\bar{x}_{0,\text{Po}213}$ for each channel. I subsequently used this scale factor to calculate another approximation: $\bar{x}_{0,\text{Ac}225}(\text{ch}0) \approx \frac{\bar{x}_{0,\text{Po}213}(\text{ch}0)}{\bar{x}_{0,\text{Po}213}(\text{ch}1)} \bar{x}_{0,\text{Ac}225}(\text{ch}1) = \frac{1863}{1462} \cdot 990 = 1262$.

Both approximations could then be contrasted with the derived value of the parameter for the latest channel 0 runs, where the position of the step edge for ^{225}Ac was visually clearer (see Fig. 1.2a).

Other members of the group carried out similar calculations independently, using runs with different labelling systems and visually checking that the derived parameters matched the spectra. The independently obtained values were compared and found to be consistent enough to define common initial conditions for the later fitting stages. The same procedure was then applied to other approximately constant parameters. The mean values, μ , were used as initial guess and a tolerance of $\pm 2\sigma$ was set using their observed standard deviations, σ , as summarised in Table 2.2. Parameter c , for example,

¹For Poisson-distributed counts, the standard deviation is approximately $\sqrt{N}$, so the relative statistical uncertainty is $1/\sqrt{N}$. Low-count bins therefore have larger relative uncertainties and the spectral features become less clear [6].

controls the low-energy slope of the smooth step and was generally found to be well behaved.

Channel	Labelling system	b	c	$x_{0,\text{Ac}225}$	$x_{0,\text{Po}213}$	$b_{\text{Po}213}$
Channel 0	Ab-DOTA	43.2 ± 13.4	0.69 ± 0.08	1240.4 ± 20.0	1859.5 ± 10.0	43.9 ± 9.0
	Macropa 10^{-5} M	47.5 ± 12.8	0.68 ± 0.10	1252.9 ± 13.0	1868.0 ± 11.6	41.8 ± 9.0
	Macropa 10^{-6} M	45.2 ± 12.2	0.71 ± 0.08	1251.7 ± 15.8	1866.5 ± 11.6	43.0 ± 8.4
Channel 1	Ab-DOTA	17.2 ± 12.2	1.56 ± 0.62	997.0 ± 20.4	1472.6 ± 30.2	12.1 ± 19.2
	Macropa 10^{-5} M	17.7 ± 18.4	1.68 ± 1.44	992.0 ± 34.2	1468.8 ± 20.0	17.5 ± 24.6
	Macropa 10^{-6} M	15.9 ± 8.6	1.36 ± 0.76	998.4 ± 26.8	1455.8 ± 21.2	39.3 ± 9.4

Table 2.2: Initial values and two-standard-deviation tolerance used in later fitting stages. Each entry corresponds to $\mu \pm 2\sigma$ and the lower and upper bounds were $\mu - 2\sigma$ and $\mu + 2\sigma$.

After narrowing down the bounds for these parameters, all runs were fitted again. To speed up the fitting process, a batch-fitting routine was developed in which all runs could be fitted at once with the same initial guesses and bounds, while retaining wider bounds for parameters that could not be constrained reliably and treating channel 0 and channel 1 data separately. Problematic runs could then be fitted individually. The resulting parameter estimates showed improved precision and consistency, and graphs of the labelled-to-total ^{225}Ac fraction like the one in Fig. 2.3 were produced.

Writing A_F and A_L for the fitted free and labelled activities and $A_T = A_F + A_L$, the free-to-total and labelled-to-total fractions were

$$R_F = \frac{A_F}{A_T}, \quad R_L = \frac{A_L}{A_T}. \quad (2.4)$$

The uncertainties shown in the ratio plots were obtained by propagating the fitted activity uncertainties from the two channels. Treating the channel uncertainties as independent, first-order uncertainty propagation gives

$$\sigma_{R_i} = \sqrt{\left(\frac{\partial R_i}{\partial A_F} \sigma_F\right)^2 + \left(\frac{\partial R_i}{\partial A_L} \sigma_L\right)^2}. \quad (2.5)$$

For the labelled-to-total fraction plotted below, this becomes

$$\sigma_{R_L} = \sqrt{\left(\frac{A_F}{A_T^2} \sigma_L\right)^2 + \left(\frac{A_L}{A_T^2} \sigma_F\right)^2}. \quad (2.6)$$

The same procedure was applied to the complementary free-to-total fraction. These propagated values account for the uncertainties from the spectral fits, but not for additional experimental uncertainties from the TLC procedure.

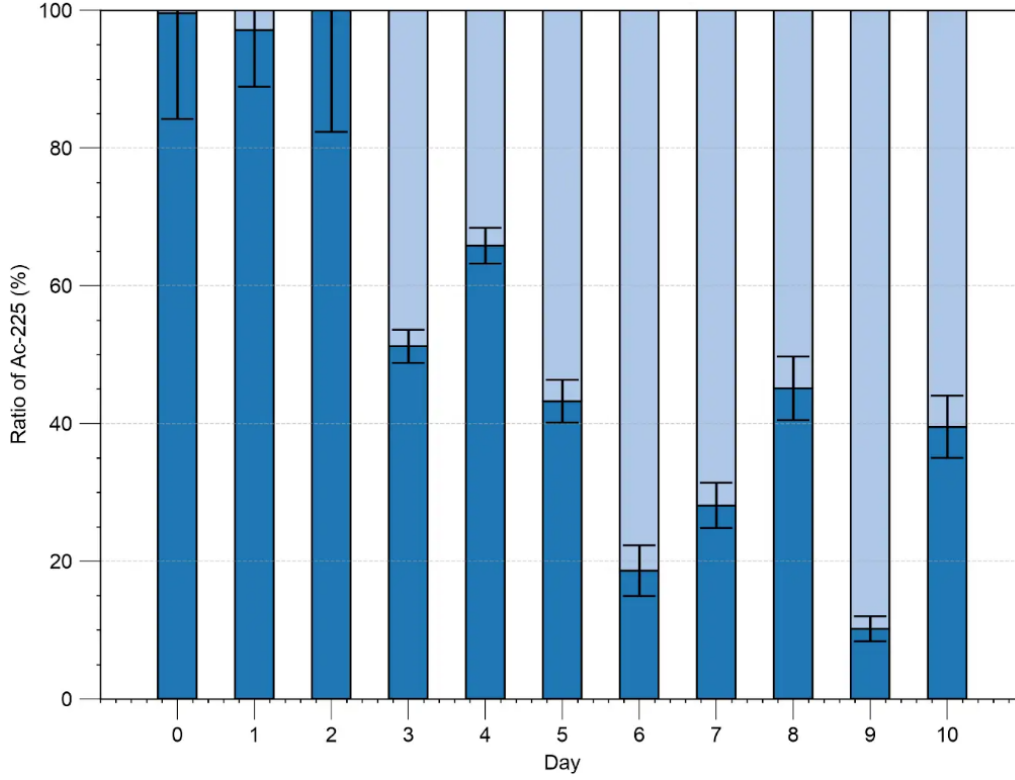


Figure 2.3: Bar graph of the labelled-to-total ^{225}Ac fraction for each day of the experiment. Measurements taken with Ab-DOTA 30 min after the start of radiolabelling.

Some fitted quantities, however, were still problematic. Specifically, parameter b was well behaved in channel 1 data, where the peaks were clearer and narrower, but varied substantially in channel 0 data and often acquired very large uncertainties. Parameter m , which controls the exponent of the power-law tail of the Crystal Ball component, spanned several orders of magnitude and reached values as large as 2.98×10^4 in some fits. Changing m by orders of magnitude sometimes had little visible effect on the fit. These observations indicated that some parameters remained poorly constrained even after the bounds had been tightened.

Given the number of free parameters and their effect on the fitting curve, I suspected that the fit could compensate for a change in one parameter through a change in another. After discussing this possibility with the group, we wondered whether a lower smooth-step amplitude, a_i , could be compensated by a higher Crystal Ball amplitude, $a_{\text{cb},i}$, effectively changing the measured channel 1 activity estimate. To investigate this, I calculated the correlation matrix of all fitted variables, with each coefficient defined as:

$$\rho_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} \quad (2.7)$$

The covariance matrix returned by SciPy's `curve_fit` function contains the estimated covariances of the fitted parameters, from which the correlation coefficients were calculated [7]. A colour scheme was then added and the result is illustrated in Fig. 2.4.



Figure 2.4: Correlation matrix of all free parameters used in one representative fit.

The colour scale makes it easy to identify that, as expected, the smooth-step amplitude was very strongly anticorrelated with the Crystal Ball amplitude of the corresponding isotope. For ^{213}Po in the representative fit shown, a_{Po213} and $a_{\text{cb,Po213}}$ had $\rho \simeq -0.99$. This indicates a strong trade-off in the parameter estimation: different combinations of the two amplitudes can produce similar spectral shapes, rather than a deterministic physical relation between them. The exact coefficients changed slightly from run to run, but the correlations nevertheless remained consistent. Additional correlations involving β_i , w_i and m_i were also identified and associated with the similar effects of other parameters in the fitting function.

To visualise these effects further, I created a Desmos graph of the fitting function in which the parameters could be varied in real time. The graph confirmed that different parameter combinations could produce similar changes in the model and made the non-linear behaviour of several parameters more apparent, particularly where changing them outside a given range had little visible effect on the fitting function.

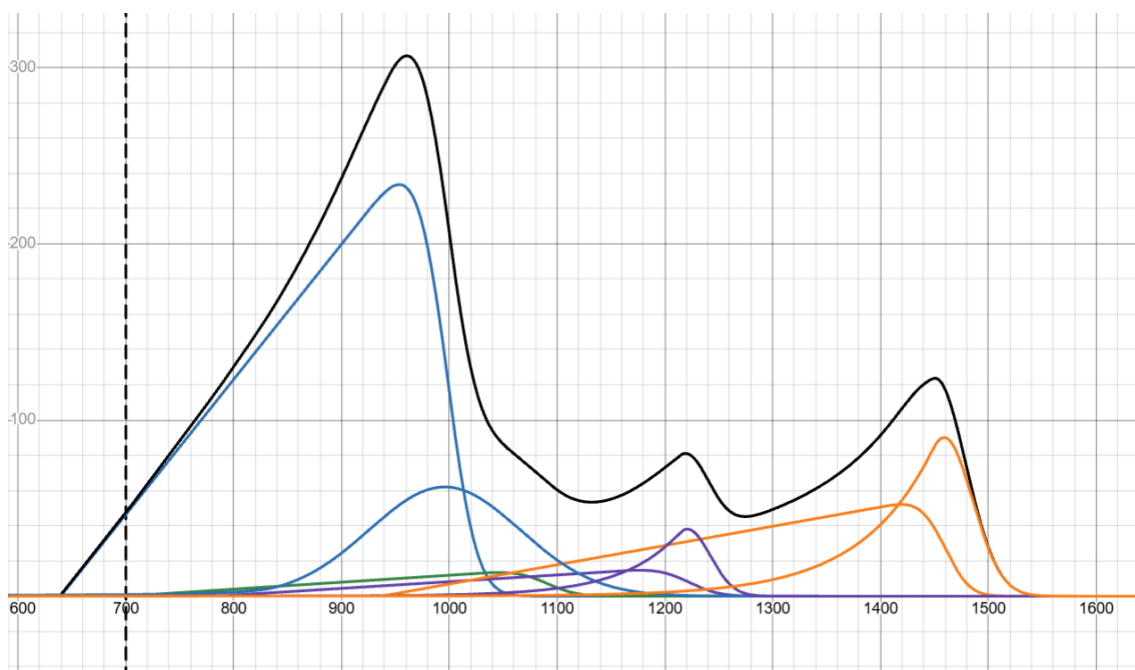


Figure 2.5: Graph of the fitting function (black) with Crystal Ball parameters for ^{225}Ac , ^{217}At and ^{213}Po on, showing its construction from smooth step and Crystal Ball components. Generated with Desmos.

For example, the higher parameter b was, the smaller the effect of varying it on the function. Varying m above values of the order of 10 had virtually no effect on the function. Parameters were therefore constrained to a range where their effect was significant. Additionally, although a_{cb} was responsible for the amplitude of the Crystal Ball peaks, β influenced the amplitude as well. Other correlations were also checked, effectively matching the information from the correlation matrix.

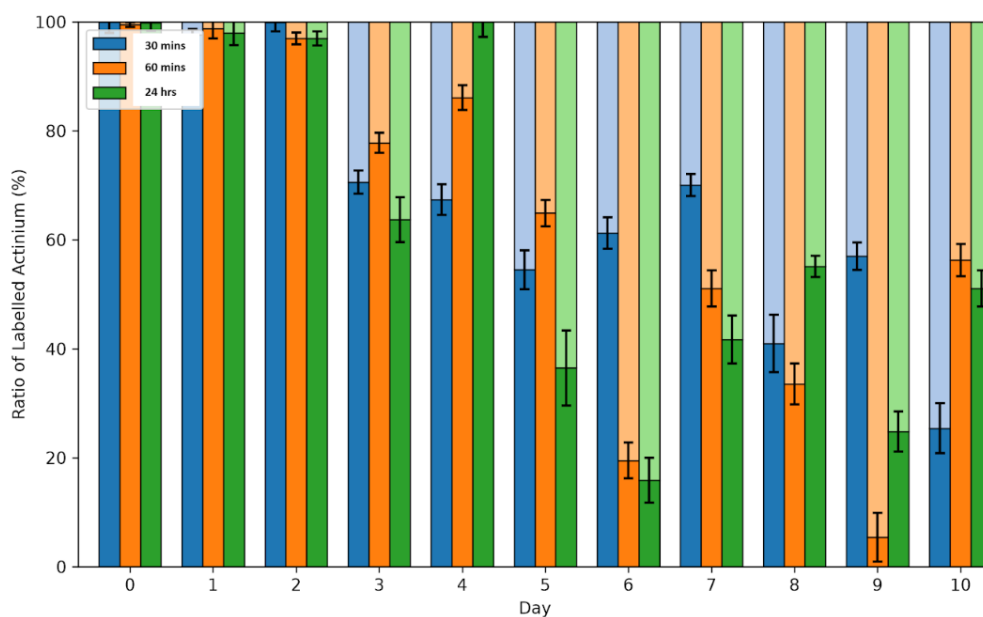


Figure 2.6: Bar graph of the labelled-to-total ^{225}Ac fraction for each day, measured 30 min, 60 min and 24 h after the start of radiolabelling with Ab-DOTA.

Upon identifying some inconsistencies in the resulting activity ratios, an incorrect numbering of some runs was identified. This was discussed with the group and corrected accordingly. A subsequent fit of all runs then yielded the graph shown above (Fig. 2.6).

No simple monotonic behaviour is apparent across the 10 study days, and the data show substantial day-to-day variability. The plotted uncertainties describe the uncertainty derived from the spectral fits and do not include the full experimental uncertainty associated with the TLC procedure. For that reason, these results require further study before firm conclusions can be drawn. Ratios between the activities of different isotopes were also monitored. This is relevant because alpha-decay recoil can release daughter nuclides from the chelator, after which they may redistribute in vivo [2]. Thus, comparing the fractions associated with the labelled component provides relevant QC information.

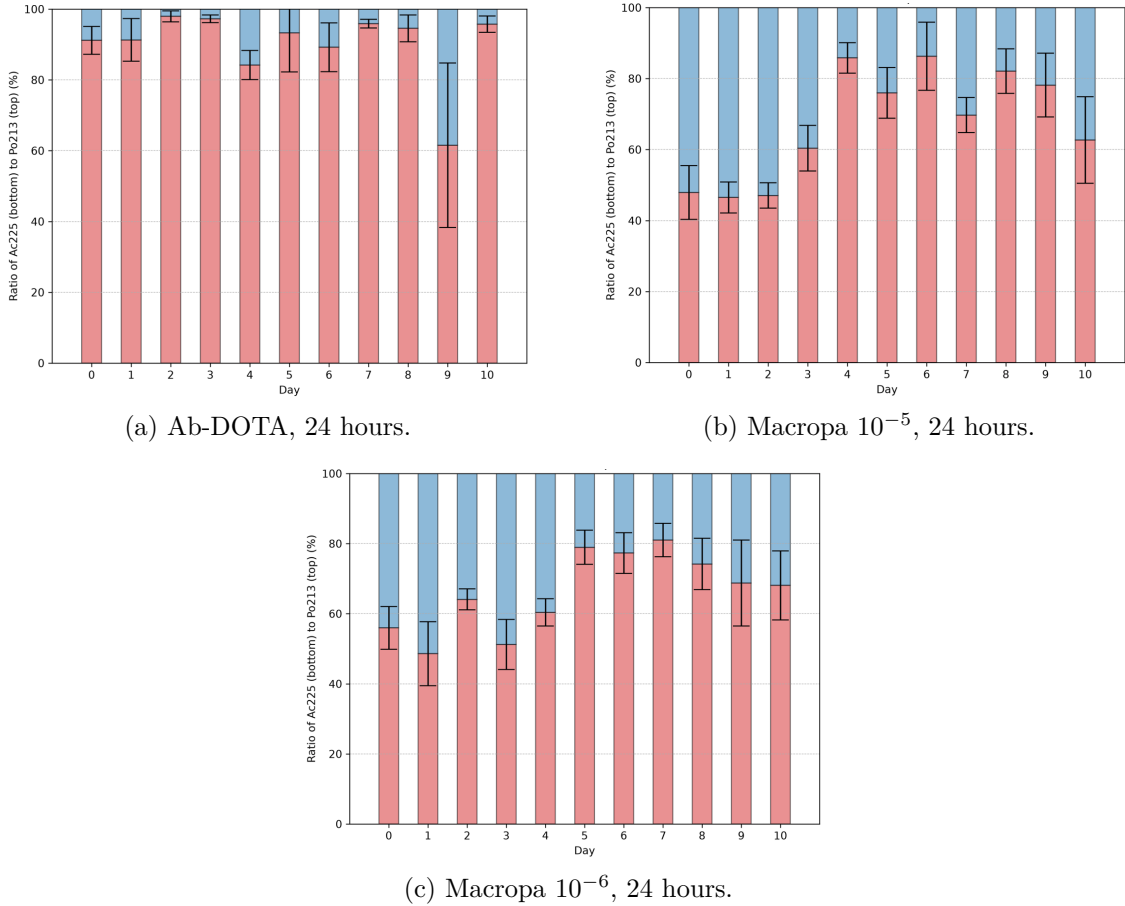


Figure 2.7: Ratios of ^{225}Ac to ^{213}Po in channel 1, corresponding to the labelled component of the solution.

Fig. 2.7 shows some of these plotted ratios for the different labelling systems. With further improvements in the fitting methodology, such comparisons could help assess their relative radiolabelling performance and stability under the tested conditions.

2.3 Unlabelled ^{225}Ac dataset

Alongside the previous task, a new unlabelled ^{225}Ac dataset from TRIUMF was analysed. It was acquired using TLC as well, but in this case no free-versus-labelled radiochemical

separation was being studied. The spectra still contained contributions from the ^{225}Ac decay chain. I started by fitting a handful of runs recorded over 48 h. As the mean number of counts was much higher than in the previous study, the positions of the step edges and other features of the spectrum were much clearer (see Fig. 2.8).

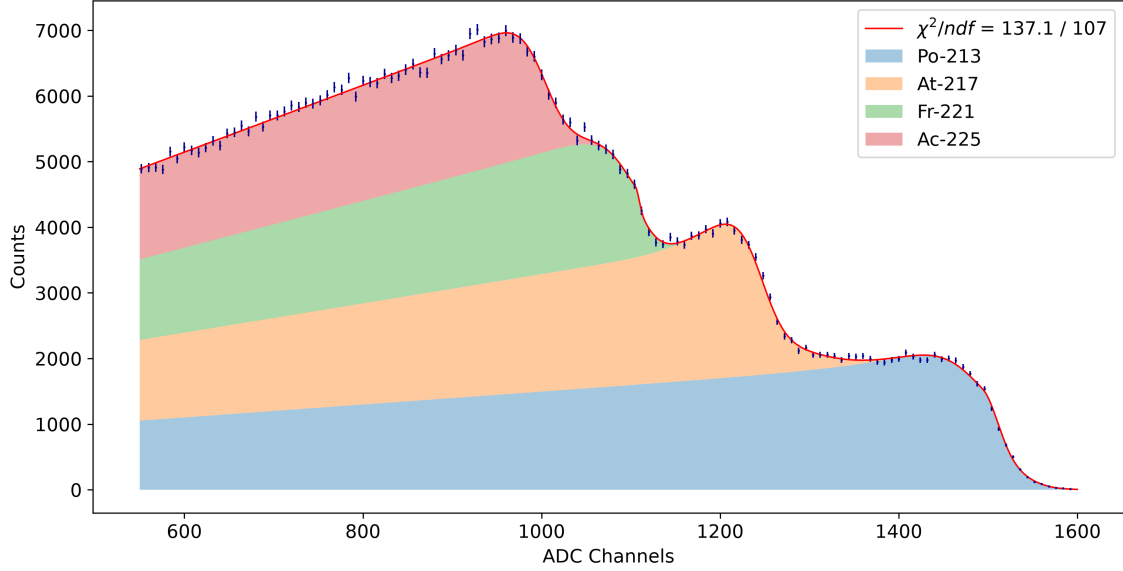
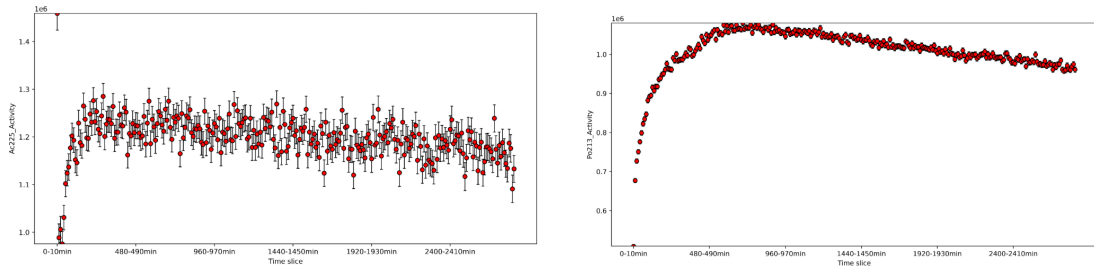


Figure 2.8: Fit of a spectrum from the more recent data showing a higher mean number of counts.

The longer acquisitions were then divided into 10 min time slices, which were all fitted using initial conditions derived from these initial fits. Some of the results are shown in Fig. 2.9.



(a) Fitted $x_{0,\text{Ac}225}$ as a function of time slice. (b) Fitted $x_{0,\text{Po}213}$ as a function of time slice.

Figure 2.9: Fitted values of $x_{0,\text{Ac}225}$ and $x_{0,\text{Po}213}$ with their uncertainties as a function of time slice. Results from the unlabelled ^{225}Ac dataset.

These results were further developed and interpreted by the rest of the group.

2.4 Toy Monte Carlo validation

Having obtained large uncertainties and apparently unreliable results in some of the previous fits, I decided to perform a simple toy Monte Carlo experiment to test the accuracy and precision of the fitting routine. I used Python to define the fitting function and evaluate it at x -positions separated by 8 ADC units, matching the bin spacing used in the fitting

analysis. For each bin i , the model prediction was used as the Poisson expectation value λ_i , and a simulated count $N_i \sim \text{Poisson}(\lambda_i)$ was generated. The function was defined using parameters derived from some of the best fits from real data. The result is shown in Fig. 2.10.

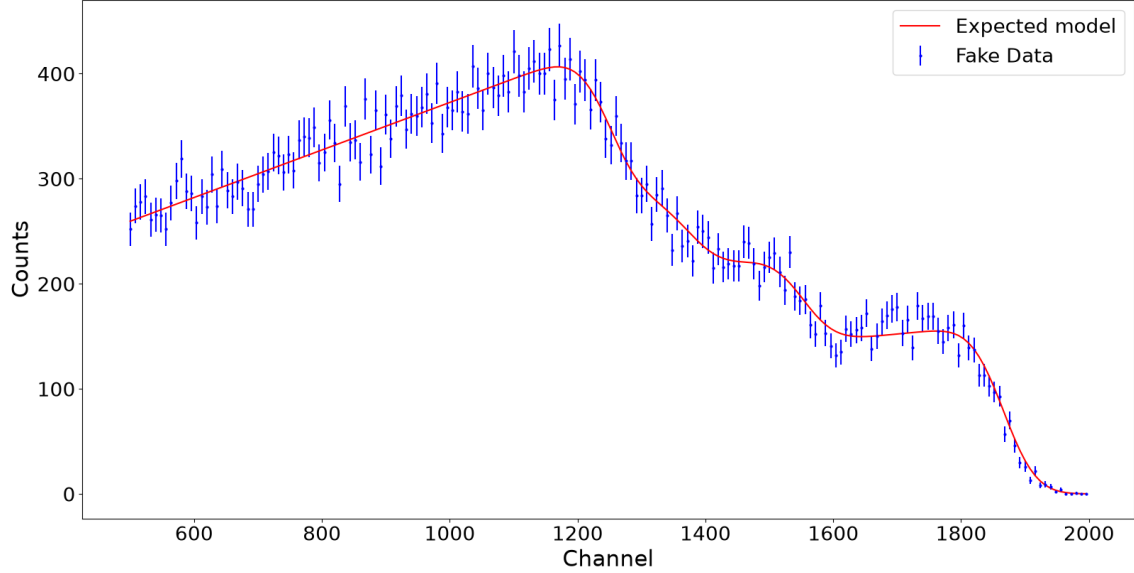


Figure 2.10: Spectrum produced from fake data generated using a Poisson distribution.

The test started without any Crystal Ball component. Because the pseudo-data were generated from the same model that was subsequently used to fit them, this study acts as a closure test of the fitting procedure rather than a validation that the model fully describes the physical spectra. Using the same procedure, 500 pseudo-datasets were generated under identical conditions. These were fitted using the batch-fitting routine. Several statistical quantities were then extracted from the ensemble to estimate how accurately and precisely the known parameters could be recovered. They are all summarised in Table 2.3.

One of the most important quantities here was the pull, which is defined as:

$$p_\theta = \frac{\theta_i - \theta_{\text{true}}}{s_{\theta,i}}, \quad (2.8)$$

where θ_i is the recovered parameter, θ_{true} is its true value and s_θ is the uncertainty reported by the fit. Repeating this for many pseudo-datasets gives a pull distribution whose mean, μ_p , should ideally be zero. A value $\mu_p < 0$ indicates a negative bias in the fitted parameter, while $\mu_p > 0$ indicates a positive bias. Similarly, the standard deviation of the pull, σ_p , should ideally be close to one. A value $\sigma_p > 1$ means the fitted uncertainties are too small relative to the observed ensemble spread, whereas $\sigma_p < 1$ indicates that they are too large [8, 9].

Each toy experiment with different initial conditions based on real data generated a table like Table 2.3.

Parameter	True value	Mean	Rel. bias (%)	Rel. std. (%)	Mean pull	Pull standard deviation
b	40.2	40.5	0.893	14.4	-0.0705	1.05
c	0.697	0.697	0.000143	2.92	0.0185	0.968
$x_{0,\text{Ac225}}$	1265	1262	-0.242	0.404	-0.510	0.871
a_{Ac225}	161	162	0.582	4.91	0.101	0.973
$x_{0,\text{Po213}}$	1865	1861	-0.215	0.112	-1.83	0.941
a_{Po213}	162	161	-0.834	1.54	-0.536	0.983

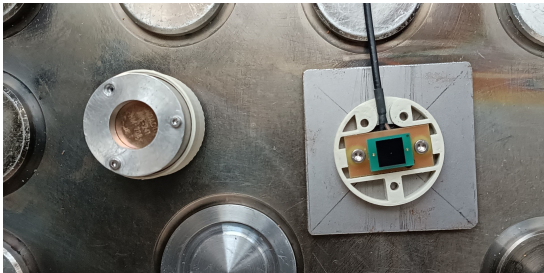
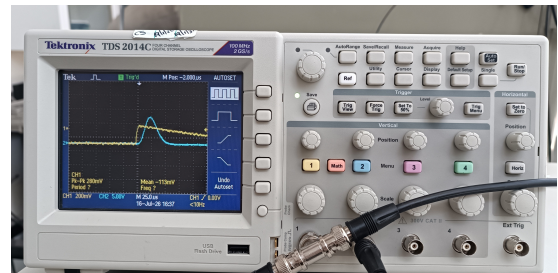
Table 2.3: Statistical results obtained from the fitting of the simulated datasets.

The relative bias was calculated as $(\bar{\theta} - \theta_{\text{true}})/\theta_{\text{true}}$, and the relative standard deviation as $\sigma_{\theta}/\theta_{\text{true}}$. We can see that the fit has difficulty determining some parameters precisely, particularly b . Its relative standard deviation is 14.4%, meaning that it varies substantially from fit to fit even though the mean recovered value is close to the true value. By contrast, $x_{0,\text{Ac225}}$ and $x_{0,\text{Po213}}$ have small ensemble spreads but negative mean pulls, indicating bias towards lower fitted values. The mean pull of $x_{0,\text{Po213}}$, -1.83 , is particularly notable because the bias is large compared with the uncertainty reported by each fit. The pull width for $x_{0,\text{Ac225}}$ is below one, indicating that its fitted uncertainties are overestimated relative to the observed ensemble spread. Similar qualitative behaviour was observed in several additional toy configurations using different true parameters and isotope activity ratios.

The mean total number of counts, summed over all bins, was also included in the calculations for each set of pseudo-spectra. A systematic study relating this quantity to the resulting precision and bias was not completed during the project and is therefore left for future work.

2.5 Detector testing

Alongside the previous analysis, we carried out experimental work in the lab to evaluate the operation of silicon detectors used to record alpha spectra. Several silicon photodiodes were connected directly to a preamplifier and placed under a radioactive ^{241}Am source (see Fig. 2.11a). The detector–source setup was placed inside a vacuum chamber, which was pumped down to approximately 1×10^{-3} mbar to reduce attenuation of the alpha particles. A reverse-bias magnitude of 19 V was used and all spectra were recorded for 5 min [3].

(a) Connected silicon detector next to the ^{241}Am source on top of its source holder.

(b) Oscilloscope displaying the raw, preamplified pulse (yellow), and the shaped pulse (blue).

Figure 2.11: Pictures of the lab setup.

The preamplifier was connected to a shaping amplifier, whose processed pulses were read out at a digitiser and translated to ADC counts (see Fig. 2.11b). The software package MAESTRO was used to produce and visualise the spectra generated with the digitiser, yielding results as seen in Fig. 2.12.

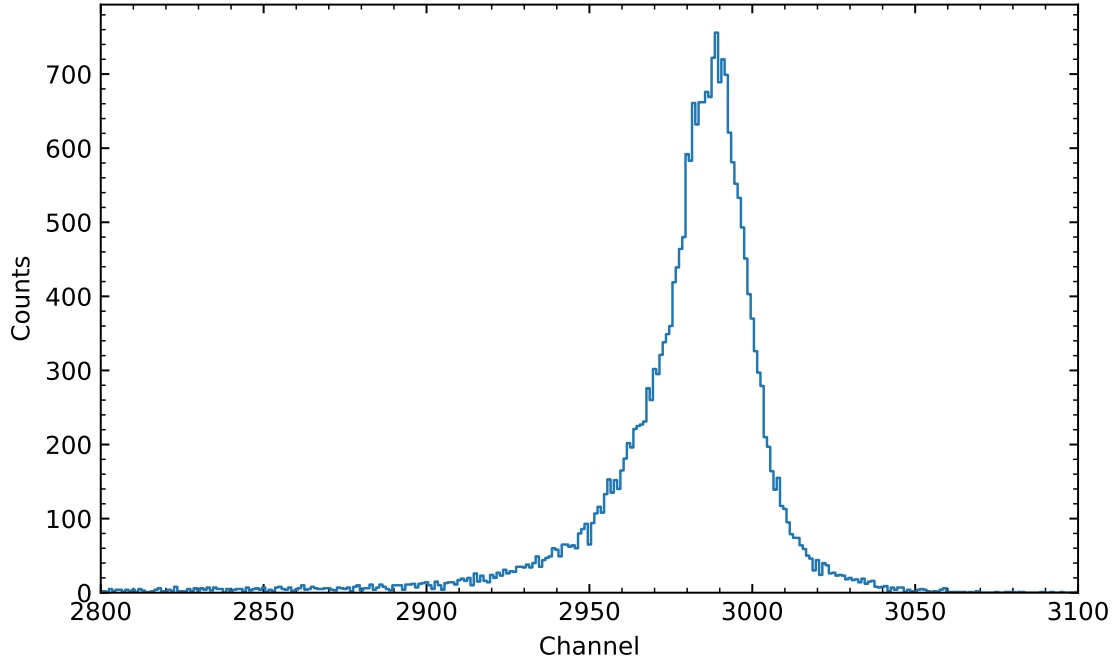


Figure 2.12: Example of a spectrum from a ^{241}Am source taken in the lab with a silicon photodiode.

An additional test was carried out to check that reversing the photodiode connection while also reversing the sign of the bias voltage did not significantly alter the resulting spectra. Eight spectra were recorded: four bias-voltage magnitudes, 5 V, 10 V, 15 V and 19 V, for each of the two connection polarities. A ^{210}Po source was used for this test. The preamplifier had been changed from the ^{241}Am measurements, so the amplifier gain and resulting ADC scale were different; comparisons were therefore made only within the ^{210}Po series. One pair is shown in Fig. 2.13.

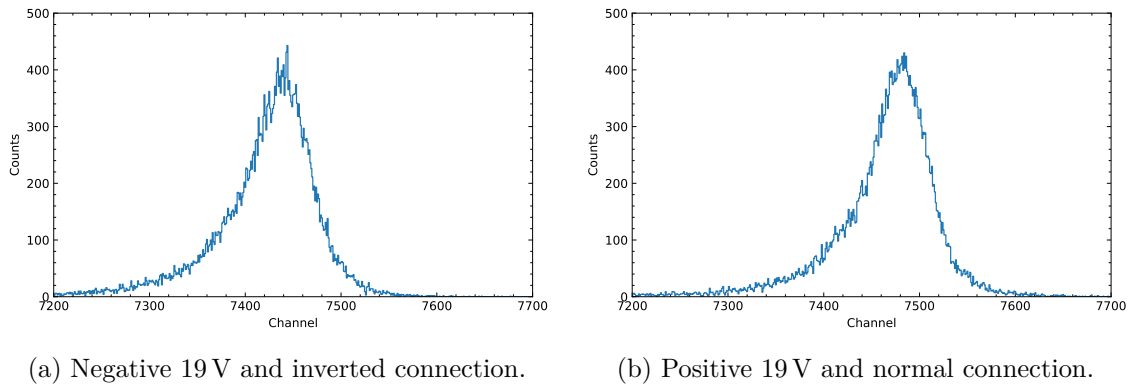


Figure 2.13: Spectra taken with a bias voltage magnitude of 19 V and a ^{210}Po source.

No substantial difference in the peak position or overall spectral shape was observed

between the two polarity configurations at the same bias-voltage magnitude. Across the full ^{210}Po series, the peak position shifted towards higher ADC channels as the magnitude of the bias increased. This behaviour is consistent with improved charge collection over the tested voltage range.

Lastly, an additional recording was taken using a ^{210}Po source, this time with a silicon detector that had had liquid from a microfluidic alpha chip spilled onto it during a previous experiment. Some parts of this detector were thought to be corroded, so we recorded another spectrum to examine its response.

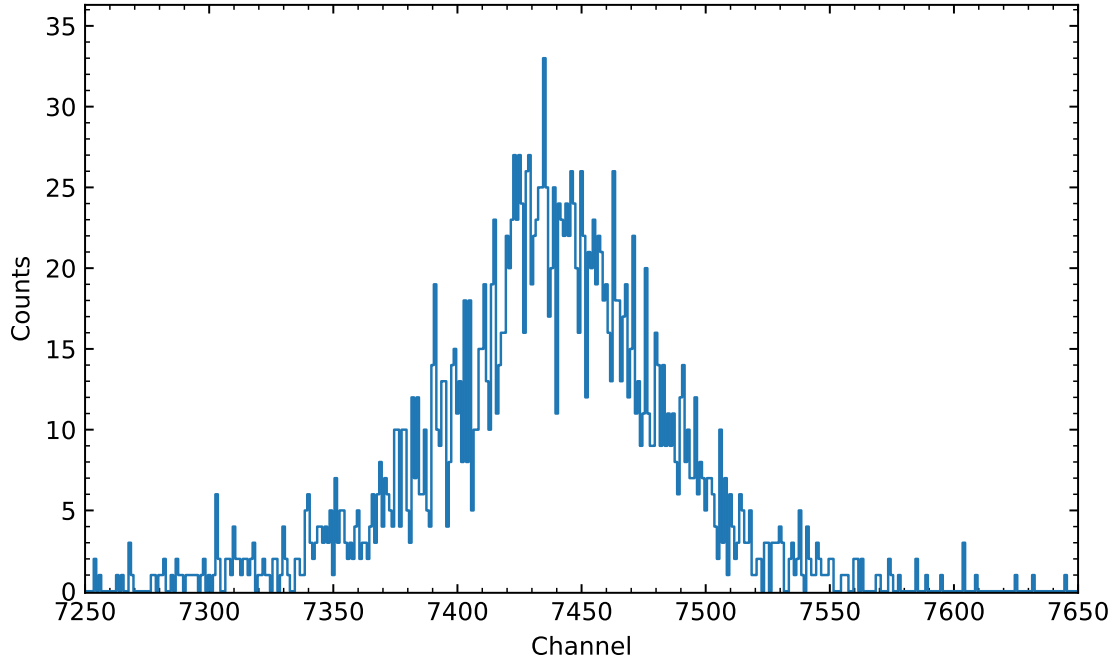


Figure 2.14: Spectrum resulting from the possibly damaged silicon detector taken with the ^{210}Po source.

As Fig. 2.14 illustrates, the spectrum had a substantially lower peak height and count level than the previous ^{210}Po spectra recorded under the same acquisition conditions. This is consistent with a degraded detector response, although no further diagnostic analysis was performed and the measurement alone cannot establish corrosion as the cause.

Chapter 3

Conclusions and future work

3.1 Conclusions

The most extensive part of this project was the analysis and optimisation of the existing fitting framework, primarily using data from the 10-day study and later applying the same approach to an unlabelled ^{225}Ac dataset. Tighter empirical bounds were introduced for several approximately constant parameters using preliminary fits and well-resolved spectra. This improved the stability, precision and consistency of the resulting fits and of derived quantities such as the labelled-to-total ^{225}Ac fraction and ratios between isotope activities. The 10-day activity ratios nevertheless showed substantial variability, and solid conclusions cannot be drawn yet.

The analysis of parameter correlations showed that some fitted quantities remained difficult to determine independently because different parameter combinations could produce similar changes in the fitting curve. I used the correlation matrix and an interactive representation of the model to identify and visualise these degeneracies.

The toy Monte Carlo validation separated several statistical limitations of the fitting procedure. Parameter b showed comparatively poor precision because of its large ensemble spread, whereas $x_{0,\text{Ac}225}$ and $x_{0,\text{Po}213}$ were determined more precisely but showed a bias towards lower values. Preliminary comparisons of configurations with more counts suggested smaller spread and bias. The pull distributions also indicated that the fitted uncertainties were not perfectly calibrated for every parameter, with some being overestimated relative to the observed ensemble spread. The simulation therefore provided separate measures of fit precision, estimator bias and uncertainty calibration.

Lastly, several nominal silicon photodiode detectors were tested in the lab with normal and reversed connection polarities and behaved as expected. Within the ^{210}Po series, increasing the magnitude of the reverse-bias voltage shifted the peak position towards higher ADC channels. A potentially damaged detector produced an anomalously low-count spectrum, consistent with a degraded detector response.

3.2 Future improvements

This project requires numerous aspects to be studied, so it has not been possible to fully develop many of the aspects covered in this report during my stay. A selection of possible

analysis tasks is summarised below.

The results obtained in Section 2.2 are not definitive and further study would be required to arrive at conclusive results. One way of proceeding would be to test how the fit, the uncertainties and the correlations react to some parameters being fully constrained to a single value.

The most relevant parameters to constrain further are the step-edge positions, x_0 . The underlying alpha-emission energies are known, and for spectra recorded with the same detector and ADC scaling, the corresponding fitted x_0 values should remain consistent. Their precise determination is nevertheless affected by the ADC calibration and by energy loss before the alpha particles reach the detector. Even though the ADC scale of the unlabelled dataset is different, the higher number of counts in these data could be useful to improve the determination of the relevant edge positions after applying the appropriate calibration.

Regarding the small toy Monte Carlo experiment, further study could quantify the effect of increasing the number of counts on the accuracy and precision of the fitting routine. The mean total number of counts has already been included in the toy calculations, so a future study could relate it systematically to parameter spread, bias and pull behaviour. The origin of the observed bias and uncertainty overestimation should then be investigated. A Poisson-likelihood fit could also be tested as an alternative to the weighted least-squares treatment, particularly for spectra containing low-count bins. If this study yielded conclusive results, it could be extended to include Crystal Ball components.

Lastly, the silicon detector spectra could be examined in greater detail to quantify their resolution and response to different bias voltages, if necessary.

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